

MEDICAL DEVICE MALFUNCTION REPORT

Report Information

Report Reference: DM-2025-0312

Date: 2025-09-01

Facility: Metro General Hospital

Department: Emergency Medicine

Reporter: Sarah Kim, Biomedical Engineer

Device Information

Device: VitaCheck Pulse Oximeter

Model: PX-500

Manufacturer: MedWatch Systems Inc.

Serial Number: PX500-2024-3847

Purchase Date: 2025-02-15

Last Calibration: 2025-05-20

Malfunction Description

During routine use in the Emergency Department, the pulse oximeter displayed SpO2 reading of 98% for a patient who was subsequently found to have SpO2 of 89% on arterial blood gas.

The device was used on a patient with:

- Dark skin pigmentation
- Poor peripheral perfusion (cold extremities)
- Hypothermia (core temp 34.2C)

The reading remained at 98% for 15 minutes until the device was repositioned to a different finger, where it displayed 91%.

Patient Impact

The delayed accurate reading resulted in a 12-minute delay in

initiating supplemental oxygen therapy. Patient recovered without permanent harm. Incident reported to Risk Management.

Patient condition at time of incident: Stable but hypoxic.

Final outcome: Full recovery after treatment.

Investigation

Preliminary findings:

1. Device was functioning within manufacturer specifications
2. Low perfusion index (0.3%) below device accuracy threshold
3. Manufacturer documentation states accuracy +/-5% when perfusion index <0.4%
4. Staff training did not cover perfusion index limitations

Corrective Actions:

1. Issue clinical advisory on device limitations
2. Update staff training to include perfusion index monitoring
3. Consider alternative devices for low-perfusion patients
4. Report to FDA MedWatch as required